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AI-Adoption, Globalization, and Firms' Labor Investment Efficiency:

Evidence From Firms in Gulf Cooperation Council Countries

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ABSTRACT

This paper examines how AI adoption affects firms' labor investment efficiency, focusing on moderating the impact of globalization in GCC nations. Using firm-level panel data from 2015 to 2024, it applies a System Generalized Method of Moments (GMM) to account for endogeneity and dynamic adjustments. Dynamic Capabilities Theory (DCT) suggests that AI adoption is a strategic resource for a firm to improve efficiency. Using four country-level globalization proxies: the Globalization Index (KOF), the Digital Adoption Index (DAI), the Global Competitiveness Index (GDI), and Total Factor Productivity (TFP), it found that globalization moderates AI adoption's impact on labor investment efficiency, supporting Globalization and Labor Market Theory, which states that globalization affects enterprises' resource optimization through global markets and technologies. It also shows that AI adoption improves labor investment efficiency significantly, and globalization's moderating effect strengthens this relationship, especially in more globally integrated countries.

KEYWORDS

AI Adoption, Globalization, Firms' Labor Investment Efficiency, GCC Economies, Dynamic Capabilities Theory (DCT), Globalization and Labor Market Theory

INTRODUCTION

Artificial intelligence (AI) transforms global corporations, raises productivity (Yigitcanlar et al., 2023), and enhances service delivery (Senadheera et al., 2025) and knowledge sharing (Li et al., 2025). It automates occupations, enhances decision-making, and optimizes workforce allocation

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(Brynjolfsson & McAfee, 2014, 2017). While previous research has demonstrated that AI enhances operational efficiency in developed countries, direct relationships between AI usage and labor investment efficiency remain scarce in emerging economies. Firms in developing countries face unique challenges, including skills gaps, infrastructure constraints, and reliance on expatriate labor (Mer & Virdi, 2024). Take robots as an example. Most developing countries are at the early stage of industrial intelligence at present; thus, companies often rely on overseas technologies. Due to technological blockades and lags, developing countries fail to access advanced robots and maintain the stability of their robot systems. Moreover, limited technology and human capital often result in inadequate operator training (Yang et al., 2022). All these imply that AI adoption's impact on developing countries may not be the same as in the developed world.

The Gulf Cooperation Council (GCC) countries, Saudi Arabia, the United Arab Emirates, Qatar, Kuwait, Oman, and Bahrain, offer an attractive environment to use AI. Saudi Vision 2030 encourages firms in these countries to rely less on the oil industry as an economic driver and transition to a knowledge economy instead (Al Naimi, 2022; Ben Hassen, 2022; Khan et al., 2020). Many industries in the GCC have followed suit. In recent research of 2,912 nonfinancial firms, companies that use more AI had higher integrated financial reporting effectiveness, especially those that hire specialized auditors with increased audit fees(Khan et al., 2024).

Most AI research has focused on developed countries. Literature that throws light on how AI can improve labor investment efficiency in the GCC is scarce (Ashfaq & Ayub, 2021; El-Saharty et al., 2020; Heath, 2021). While AI raises business productivity (Ardito et al., 2024; Brynjolfsson et al., 2021), rapid economic development in the GCC may raise AI usage and enhance labor efficiency. For example, audit automation enhances auditing efficiency (Khan et al., 2022). AI systems adapt to learning, narrow the skill gap in developing economies, and improve labor force quality (Huang & Lin, 2024; Wang et al., 2024). However, lack of sufficient technical infrastructure, skills mismatches, and cultural unwillingness to change might slow AI usage (Hradecky et al., 2022). Thus, conducting research allows us to understand the impact of these limitations on AI adoption and labor investment efficiency.

Much of the previous research has focused on AI as a driver of productivity, but little research has examined the relationship between AI adoption and labor investment efficiency in emerging nations. Ben-Nasr and Alshwer (2016) stress the necessity of aligning labor investments with company goals, although the role of AI in emerging economies remains unclear. AI adoption allows GCC enterprises to optimize labor allocation, especially in digitally transformed industries (Khan et al., 2022). AI adoption and labor investment efficiency have been researched individually; however, the moderating role of globalization has not been studied, to the best of our knowledge. Globalization strengthens information exchange networks and enables firms to access modern technologies and a worldwide talent pool, which influences AI adoption as a result (Dauvergne, 2021). In a globalized economy, firms can rapidly adopt new technologies owing to foreign expertise and increased competition (Skare & Riberio Soriano, 2021). The literature regarding how globalization affects AI adoption and labor investment efficiency is sparse (Sala & Trivín, 2018; Skare). The Globalization Index (KOF), the Digital Adoption Index (DAI), the Global Competitiveness Index (GCI), and total factor productivity (TFP) are used to examine the impact of globalization on AI usage and labor investment efficiency.

Our research has several contributions. First, it examines the direct relationship between AI adoption and labor investment efficiency in GCC firms. Globalization moderates the uptake of AI and labor efficiency. This research contributes to the literature with 3,413 GCC firm-year observations spanning 2016–2024. Panel data analysis and system Generalized Method of Moments (GMM) is used to evaluate how AI adoption impacts labor efficiency while accounting for endogeneity. The research employs various proxies for globalization to demonstrate how it influences AI usage and its impact on developing nations.

LITERATURE REVIEW

The dynamic capabilities and globalization and labor market theories explain the relationship between AI usage and labor investment efficiency. Dynamic capabilities and globalization theory (Teece et al., 1997) proposed that companies achieve a competitive advantage by implementing and adapting internal competencies to meet the technological and market changes. Automation, data-driven decision-making, and predictive workforce management enhance labor investment efficiency by using AI (Johnson & Reyes, 2021). AI increases productivity, minimizes inefficiencies, and optimizes labor allocation in real time (Teece et al.,1997). Companies rely on AI's benefits to hire, teach, and retain staff that can meet their market needs and budget requirements (Jaiswal et al., 2023), avoid hiring excess or underutilized staff, improve labor investment efficiency, and enhance human capital management (Riberio Soriano, 2021).

HYPOTHESIS DEVELOPMENT

AI and Firms' Labor Investment Efficiency

AI empowers companies to enhance staff capacities and address skill mismatches (Jaiswal et al., 2023) by utilizing algorithms to pinpoint employee skill gaps and offer personalized training (Kgoale, 2023). It helps firms adapt to technological and market changes by ensuring that their employees match the ever-changing company needs, thereby enhancing labor efficiency. AI-driven skill optimization programs can improve labor productivity, particularly in industries that have adopted digital transformation (Davenport & Ronanki, 2018). AI can target recruitment to ensure business organizations select the right candidates and enhance labor investment efficiency by providing staff that supports strategic goals (Birkinshaw, 2020; Brynjolfsson & McAfee, 2014). AI adoption and labor investment efficiency may differ in the GCC due to the region's unique labor market characteristics. The GCC region's substantial reliance on expatriate labor and ongoing economic diversification projects, such as Saudi Vision 2030, make it an ideal place to investigate how AI can enhance labor efficiency (Zeren, 2025). The region's rapid economic and technological changes, along with its drive to modernize the workforce, enable diversification, making this research particularly crucial. These arguments led us to the following hypothesis:

Hypothesis 1: AI adoption enhances firm labor investment efficiency in the GCC region.

Moderation Effect

Previous studies researched the KOF to measure a nation's economic, social, and political integration into global markets, i.e., the degree of globalization on digital penetration using the DAI, the GCI, and the TFP as variables in the panel model. Globalization presents a new opportunity to circulate information, with a positive impact on technology spillovers and transfers (Skare & Riberio Soriano, 2021). On the other hand, institutional restrictions and restricted access to the best AI applications reduce the link between AI usage and labor investment efficiency in less globalized companies (Abbas Khan et al., 2024). Thus, this research hypothesizes that:

Hypothesis 2: Globalization affects the benefits of AI use. Companies in countries with higher globalization, digital adoption, competitiveness, and TFP enjoy more efficient labor investment gains due to AI adoption.

DATA, VARIABLES, AND METHODS

Data

The data include company websites, financial databases, and government-regulated portals that archive sustainability and annual reports. Since multinational enterprises (MNEs) in the region were required to incorporate sustainability reporting in their annual reports starting in 2015 (Topple et al., 2017), the period from 2015 to 2024 was selected for the analysis of diversity management construction. This method produced a large sample of 354 MNEs (162 local and 192 foreign MNEs) from around the GCC (Santomartino et al., 2022), assuring the study's relevance and completeness. We differentiate between local and foreign firms based on geographic origin and ownership/control. Foreign MNEs have foreign headquarters or significant operations as well as foreign ownership or control. Local MNEs, even if they operate worldwide, are owned or controlled by domestic stakeholders where their headquarters have significant operations in the country. The company headquarters and ownership impose differences between the two. Table 1 presents year-wise data for both foreign and local MNEs. For a valid analysis, we included firms with at least five years of data. This selection criterion accounts for long-term trends and changes, improving assessment accuracy and confirming our conclusions.

Table 1. Data Description

Year	Foreign MNEs	Local MNEs	Total
2015	174	131	305
2016	177	136	313
2017	185	137	322
2018	190	144	334
2019	191	158	349
2020	194	164	358
2021	194	164	358
2022	194	164	358
2023	194	164	358
2024	194	164	358
Overall observations	1,887	1,526	3,413

Note. MNE = multinational enterprise.

AI Adoption

Financial reports are important sources for analyzing AI usage, as they reveal an organization's strategic intentions and technology investments. We used advanced text analysis and annual (10-K) financial data to estimate AI use among GCC enterprises (Grover et al., 2022; Gupta & Rathore, 2024; Wang et al., 2025). This study extracted and analyzed reports from company websites and financial regulatory platforms to evaluate organizations' engagement with AI-driven technology in a robust and replicable manner. By doing so, it could eliminate selection bias. Regarding firm-level annual reports, the analysis included only publicly listed corporations with 10,000 reports to ensure data consistency and comparability. Advanced natural language processing was employed to extract insights into AI adoption from the dataset (Elahi et al., 2023; Thayyib et al., 2023). Drawing on literature and industry best practices, this study developed a tailored term lexicon to identify AI adoption. In addition to “artificial intelligence,” “machine learning,” “deep learning,” and “natural

language processing," the dictionary also included "AI-driven automation," "predictive analytics," and "data-driven decision-making." To enhance accuracy, we have made a clear distinction between organizations discussing AI as an industry trend and those actively using AI in their day-to-day operations.

Python-based natural language processing frameworks (NLTK and spaCy) were used to construct a rigorous text-processing pipeline after keyword detection (Martinc et al., 2024; Perovšek et al., 2016). Stop words were deleted, stemming and lemmatization were performed, and AI-related phrases were found in meaningful contexts (Martinc et al., 2024; Perovšek et al., 2016). Regular expressions and topic modeling also distinguished AI references from mentions of corporate strategy and investment. This refined technique avoided overestimating AI adoption by excluding comments on generic AI trends and industry advancements.

Dual-layered scoring was used to measure AI adoption (Martinc et al., 2024). First, organizations that mentioned AI in a relevant context were classified as adopters (1), whereas those without AI references were classified as non-adopters (0). Since AI adoption is a spectrum, we created a frequency-weighted metric. This metric counted AI-related phrases standardized by report word count. LogAI was used to balance adoption intensity and prevent skewness from firms with high AI mentions. We verified our method several times to boost dependability. A random sample of articles was thoroughly examined to ensure AI references were included instead of aspirations or general industry opinion. Sentiment analysis was applied to AI-related sections to determine whether firms were reporting actual AI operations or merely a trend (Martinc et al., 2024). Robustness tests decreased misclassification and improved AI adoption.

A final AI adoption measure included binary adoption status, phrase frequency weighting, and contextual relevance. This enterprise-wide AI engagement measure enabled thorough econometric analysis. This systematic approach enabled us to quantify the utilization of AI by GCC firms without the potential for self-reporting biases (Martinc et al., 2024). This technique confirmed our results and established a repeatable methodology for measuring AI adoption.

Firm Labor Investment Efficiency

Labor investment efficiency depends on the deviation of business labor investment from economic fundamentals. Net hiring, the percentage change in employee numbers, is the primary metric (Ben-Nasr & Alshwer, 2016). Labor investment efficiency may be estimated by the absolute difference between actual net hiring and the predicted level based on firm-specific economic characteristics (ABN-NETHIRE; Habib & Hasan, 2021). Labor investment distortions suggest over- and underinvestment. Lower ABN-NETHIRE values indicate more efficient labor investment as businesses adjust workforces to operational demands (Cao & Rees, 2020). The following regression (Equation 1) measures ABN-NETHIRE.

$$\begin{aligned} NETHIRE_{i,t} &= \beta_0 + \beta_1 SALESGROWTH_{i,t} + \beta_2 SALESGROWTH_{i,t-1} \\ &+ \beta_3 \Delta ROA_{i,t} + \beta_4 \Delta ROA_{i,t-1} + \beta_5 ROA_{i,t} + \beta_6 RETURN_{i,t} + \beta_7 SIZE_{i,t} \\ &+ \beta_8 LIQUIDITY_{i,t} + \beta_9 LIQUIDITY_{i,t-1} + \beta_{10} \Delta LIQUIDITY_{i,t} \\ &+ \beta_{11} LEVERAGE_{i,t} + \beta_{12} LOSS1_{i,t-1} + \beta_{13} LOSS2_{i,t-1} + \beta_{14} LOSS3_{i,t-1} \\ &+ \beta_{15} LOSS4_{i,t-1} + \beta_{16} industry\ effect_{i,t} + \beta_{17} Covid-19\ dummy_{i,t} + \varepsilon_{i,t} \end{aligned} \tag{1}$$

In Equation 1, $NETHIRE_{i,t}$ is the percentage change in the number of employees of firm i in year t; $SALESGROWTH_{i,t}$ is the percentage change in the sale of the firm i in year t; ROA is the return on assets; $RETURN_{i,t}$ is the annual stock return; $SIZE_{i,t}$ is the log of the market value of equity at the beginning of the year; $LIQUIDITY_{i,t}$ is the ratio of cash and short-term investments plus receivables to current liabilities; $LEVERAGE_{i,t}$ is the ratio of long-term debt to total assets at the

beginning of the year; $LOSS_{i,t\text{-}1}$ is an indicator variable for each 0.005 interval of prior-year *ROA* from 0 to −0.025.

Empirical Model

This study examines how AI adoption affects ABN-NETHIRE. As in the literature, it uses a robust dynamic panel two-step system-GMM estimation method to examine in the Ghanaian capital market. For dependability and advantages, the system-GMM estimator is used. To account for AI adoption (AI-ADOPTION) and ABN-NETHIRE's dynamic character, the system-GMM estimator combines difference and level equations (Semykina & Wooldridge, 2010). Endogeneity, heteroskedasticity, simultaneity, and nonlinearity in panel data with unobserved effects are handled (Semykina & Wooldridge, 2010). It uncorrelated error terms across data and used time-invariant indicators to account for unobserved variability (Blundell & Bond, 1998). Lagged dependent variables regulate endogeneity and reverse causation (Asongu & Acha-Anyi, 2019). The system-GMM model is verified by serial correlation, overidentification restrictions, and contemporaneous correlation (Asongu & Acha-Anyi, 2019). These tests demonstrate the model's statistical validity and confirm that the instrumental variables are uncorrelated with the residuals (Asongu & Acha-Anyi, 2019). A rigorous econometric technique is employed to investigate the relationship between IC and FP in GSE-listed manufacturing companies, addressing methodological issues and ensuring the reliability of the findings. The regression model in Equation 2 was utilized.

$$\begin{aligned} NETHIRE_{i,t} &= \beta_0 + \beta_1 ABN - NETHIRE_{i,t-1} + \beta_2 AI - ADOPTION_{i,t-1} \\ &+ \beta_3 TMT - CONTROL_{i,t} + \beta_4 FINANCIAL\ CONTROL_{i,t-1} + \beta_5 year\ effect_{i,t} \\ &+ \beta_6 industry\ effect_{i,t} + \beta_7 country\ effect_{i,t} + \beta_8 Covid - 19\ dummy_{i,t} + \varepsilon_{i,t} \end{aligned} \tag{2}$$

Following previous studies (Huang & Tarkom, 2022; Li et al., 2024), AB-NETHIRE is the absolute value of the difference between actual net hiring and the expected level; AI-ADOPTION is the index value created through context analysis (see the Data section). We control the top management team (TMT) composition, which has been found to significantly affect group functioning, corporate governance, and business performance. The TMT's functional background, tenure, age, gender diversity, and ethnic diversity can all impact decision-making and investment efficiency. The TMT comprises the company's top five executives, including the CEO. Organizations where the CEO is also the board chair were also considered. CEO dualism was not expected to affect the TMT measure significantly (González & Greve, 2023). We employ Blau's Index of Heterogeneity to measure TMT diversity by quantifying variety in functional background, race, and gender. The Blau index is calculated as in Equation 3.

$$Blau\ index = 1 - \sum PK2, \tag{3}$$

where *P* denotes the proportion of unit members in the *K*th category. Thus, the values of Blau's index can range from zero to $^{K\text{-}1}/_{K}$, with *K* indicating the number of potential categories (Lakshman & Gonzalez, 2023). Comparing Blau indices for gender and functional background is tricky since they measure diversity differently. The Blau index maximum for gender is 0.5 because it has only two categories for gender. In comparison, functional background has eight categories. Hence, the maximum Blau index score is 0.875 if the team represents each function equally. That each index captures different levels of diversity explains the variance in categories. A higher Blau index implies team diversity, whereas a lower score indicates homogeneity. We measured age heterogeneity within each firm using the coefficient of variation, which is the standard deviation divided by the mean team age. This approach shows the team's age diversity. Tenure diversity was determined using the coefficient of variation of team members' years of service (Porcena et al., 2021).

Labor investment efficiency and sustainability depend on financial controls, which govern firms' long-term growth and resource allocation. Firm leverage, size, and tangibility influence a firm's investment efficiency level (Habib & Hasan, 2021). Firms with high financial leverage (LEVERAGE) may be involved in high-return projects or avoid risks that affect long-term sustainability. Higher market-to-book (MTB) ratios indicate growth expectations and encourage sustainability investment to preserve competitive advantages (Habib & Hasan, 2021). Companies' reliance on physical capital, that is, tangible assets (TANGIBLES), limits flexible, innovation-driven sustainability project investment (Singh, 2023). The standard deviations of sales, cash flow, and employee volatility (SD-SALES, SD-CASH FLOW, and SD-NETHIRE) indicate a corporation's financial stability, which affects its willingness to invest in risky long-term projects. Finally, dividend distributions may show a company's focus on short-term success and discourage sustainable behaviors. All these showcase the relationship between financial health and sustainable investment decisions. Table 5 in the Appendix lists the definitions.

RESULTS

Descriptive Statistics and Correlation

In Table 2, enterprises that use AI have a mean of 0.256 and a high standard deviation (2.212), indicating significant variation. This significant discrepancy suggests that some companies utilize numerous AI tools for operations and enhance their profitability while others have minimal applications. They have an average FIRM-SIZE of 7.385 (in logarithmic terms), implying that large companies can apply AI due to their vast financial and technological resources. Their mean solid age of 6.461 years suggests the level of maturity. LEVERAGE (mean 0.248) and dividend payout (mean 0.476) indicate that AI use may not directly relate to financial risk or shareholder payout. Firms that utilize AI are younger and more diverse, implying that younger and more diverse leadership TMTs may be more open to technology. AI-adopting enterprises had slightly higher SD-SALES volatility, probably because they can respond to rapidly changing markets.

Table 2. Descriptive Statistics and VIF (Variance Inflation Factor)

Variable	Mean	Median	Standard Deviation	VIF
ABN-NETHIRE	0.256	0.244	0.182	-
OVER-LABOR	0.163	0.123	0.275	2.254
UNDER-LABOR	-0.104	-0.097	0.111	1.887
AI-ADOPTION	0.116	0.103	2.212	2.034
FIRM-SIZE	7.385	7.237	1.582	1.740
FIRM-AGE	6.461	6.206	1.595	1.593
LEVERAGE	0.248	0.210	0.251	2.327
SD-SALES	0.145	0.104	0.136	1.740
SD-CASH FLOW	0.054	0.038	0.058	1.960
SD-NETHIRE	0.178	0.111	0.238	1.593
TANGIBLES	0.289	0.212	0.238	1.813
DIVIDEND	0.476	0.444	0.506	1.887
LABOR-INTENSITY	0.006	0.003	0.011	2.36

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Table 2. Continued

Variable	Mean	Median	Standard Deviation	VIF
TMT age	44.719	41.290	4.531	2.107
TMT diversity	0.284	0.274	1.095	1.813
TMT race	0.476	0.456	0.122	1.887
TMT tenure	6.021	5.818	2.453	2.180
KOF	61.694	62.506	9.777	1.877
DAI	56.561	56.294	10.943	2.066
GCI	70.353	69.770	11.287	2.054
TFP	57.460	58.970	7.445	1.365

Note. AI = artificial intelligence; VIF = variance inflation factor; TMT = top management team; KOF = Globalization Index; DAI = Digital Adoption Index; GCI = Global Competitiveness Index; TFP = total factor productivity; ABN-NETHIRE = abnormal net hiring; SD = standard deviation; SD-NETHIRE = standard deviation of employee volatility.

According to Skare and Riberio Soriano (2021), the KOF, which measures globalizations' political, social, and economic aspects, has a mean of 64.55 and a standard deviation of 10.23, indicating modest regional globalization but significant country-level variations. The DAI measures a country's digital adoption across business, government, and individual sectors. It has a mean of 59.18 and a larger standard deviation of 11.45, indicating significant diversity in digital adoption among GCC countries due to varying digital transformation levels and market sizes. Finally, TFP, which represents the portion of output not explained by the number of inputs used in production, has a mean of 60.12 and a standard deviation of 7.79, indicating less variation in regional output than the other indices.

The variance inflation factor (VIF) for each variable was calculated to assess multicollinearity. In statistics, VIF is the ratio (quotient) of a parameter estimate's variance when fitting a complete model with additional parameters to its variance when fitting the parameter alone (Khishdari et al., 2025). Collinearity increases the variance of the estimated regression coefficient as per the VIF index (Kim, 2019). The VIF values are below 5%, indicating no multicollinearity and low predictor correlation (Dertli et al., 2024).

AI and Labor Investment Efficiency

GMM estimation-weighted statistics strengthen models. Table 3 shows a significant one-year lag in AI adoption ($p < .01$), which validates the GMM estimation (Saygın & İskenderoğlu, 2022). This lag structure accommodates endogeneity and temporal dynamics, thereby enhancing the credibility of model conclusions. Column 1 represent weighted statistics keep their value in all models. Since the independent variables explain 52.2% of the variance in the dependent variable, the R-squared score of 0.526 indicates a good model fit (Okoyeuzu et al., 2021). The adjusted R-squared value of 0.509 suggests that the number of predictors does not overstate the explanatory power (Saygın & İskenderoğlu, 2022). Saygın and İskenderoğlu (2022) found that the model's predictions match the observed data, with a low regression standard error (0.243) and a significant effect of the independent variables (F-statistic: 152.679, p-value: 0.000). The mean VIF of 1.535 does not suggest multicollinearity, as the predictor variables are poorly correlated. A considerable first-order autocorrelation ($r = 3.908$, $p < 0.01$) in the residuals suggests a time-based relationship in the data. No higher-order autocorrelation is seen in the AR(2) value of 0.224 (Haque, 2021). The Sargan overidentifying restrictions (OIR) test (0.321) and Hansen OIR test (0.277) confirm that the GMM model's instruments are valid and

that overidentification is not a concern, confirming the dependability of the findings. These statistics demonstrate a well-specified model, valid instruments, and robust findings.

Table 3 shows the baseline findings on AI-ADOPTION and ABN-NETHIRE, where ABN-NETHIRE is the absolute residual, as shown in Column 1. The computed AI-ADOPTION coefficient is positive and statistically significant ($p < .05$), signifying a strong inverse relationship between AI adoption and labor investment inefficiency. The negative coefficients show that higher AI adoption ratings are associated with fewer minor deviations in labor investment from economically justified levels, suggesting greater labor allocation efficiency. Thus, firms using AI make more accurate and efficient workforce recruiting decisions with less economic difference between actual and expected labor investment (Li et al., 2023a). The results further highlight the economic importance of AI usage: a one-standard-deviation increase in AI adoption reduces labor investment inefficiency by 9.61%. This finding demonstrates that AI adoption enhances labor allocation both statistically and practically, thereby improving decision-making and workforce management strategies (Mishra et al., 2022; Wang et al., 2024b).

In Columns 2 and 3 (Table 3), we re-estimate the baseline regression model using subsamples of firms with labor overinvestment (positive abnormal net hiring—actual net hiring exceeds expected levels) in Column 2 or labor underinvestment (negative abnormal net hiring—actual net hiring below expected levels) in Column 3. In all subsample models, we use the absolute value of abnormal net hiring (ABN-NETHIRE) as the dependent variable for clarity and consistency. A positive coefficient on AI-ADOPTION in either subsample indicates that AI-ADOPTION, as reflected in higher AI adoption scores, is associated with superior labor investment efficiency, regardless of whether the firm over- or underinvests in labor. The results indicate that organizations utilizing AI make more informed and efficient labor investment decisions. This suggests that AI technologies enhance labor investment efficiency and mitigate inefficient labor allocation in both directions.

In Column 4, the two-stage least squares supports the major findings, which remain unchanged from those reported in Column 1. The results align with those in Column 1, which strengthens our conclusions. This agreement, available in various models, supports AI usage and enhances labor investment efficiency in GCC countries.

Table 3. Artificial Intelligence (AI) Adoption and Abnormal Net Hiring (ABN-NETHIRE)

	Model 1		Model 2		Model 3		Model 4	
	ABN-NETHIRE		ABN-NETHIRE		ABN-NETHIRE		ABN-NETHIRE	
	Coefficient	S/E	Coefficient	S/E	Coefficient	S/E	Coefficient	S/E
AI-ADOPTION (H1)	0.031**	(0.015)	0.022**	(0.010)	0.019**	(0.009)	0.027**	(0.013)
One-year lagged								
(ABN-NETHIRE)	0.618***	(0.141)	0.604***	(0.138)	0.626***	(0.143)	0.618***	(0.141)
Financial control								
FIRM-SIZE	0.08***	(0.016)	0.158***	(0.032)	0.146***	(0.030)	0.107**	(0.022)
FIRM-AGE	0.056**	(0.027)	0.074**	(0.035)	0.**	(0.019)	0.113*	(0.054)
LEVERAGE	-0.102**	(0.047)	-0.121*	(0.055)	-0.082*	(0.037)	-0.141**	(0.064)
MTB	0.062*	(0.036)	0.091**	(0.053)	0.095*	(0.055)	0.089*	(0.052)
SD-SALES	-0.111**	(0.049)	-0.107**	(0.047)	-0.111**	(0.049)	-0.112**	(0.049)
SD-CASH FLOW	-0.125***	(0.023)	-0.121***	(0.022)	-0.125***	(0.023)	-0.126***	(0.023)
SD-NETHIRE	-0.022**	(0.010)	-0.021**	(0.010)	-0.022**	(0.010)	-0.020**	(0.009)
TANGIBLES	0.002	(0.002)	0.002	(0.002)	0.002	(0.002)	0.002	(0.002)

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Table 3. Continued

	Model 1		Model 2		Model 3		Model 4	
	ABN-NETHIRE		ABN-NETHIRE		ABN-NETHIRE		ABN-NETHIRE	
	Coefficient	S/E	Coefficient	S/E	Coefficient	S/E	Coefficient	S/E
DIVIDEND	0.072**	(0.306)	0.070**	(0.298)	0.072**	(0.306)	0.073**	(0.310)
LABOR-INTENSITY	0.066*	(0.035)	0.064*	(0.034)	0.066*	(0.035)	0.067*	(0.036)
TMT control								
TMT age	0.001	(0.001)	0.001	(0.001)	0.001	(0.001)	0.001	(0.001)
TMT diversity	0.032*	(0.018)	0.031*	(0.017)	0.032*	(0.018)	0.032*	(0.017)
TMT race	0.080**	(0.034)	0.077**	(0.033)	0.080**	(0.034)	0.081**	(0.035)
TMT tenure	0.091**	(0.044)	0.088**	(0.042)	0.091**	(0.044)	0.092**	(0.044)
Fixed control								
Country fixed effect	Yes		Yes		Yes		Yes	
Year fixed effect	Yes		Yes		Yes		Yes	
Industry effect	Yes		Yes		Yes		Yes	
Covid-19 effect	Yes		Yes		Yes		Yes	
Weighted statistics								
$R-$ squared	0.526		0.562		0.526		0.553	
Adjusted R squared	0.509		0.544		0.509		0.534	
Regressions' S/E	0.243		0.259		0.243		0.255	
F-statistics	152.679		163.206		152.679		160.260	
Probability (F-statistics)	0.000		0.000		0.000		0.000	
Mean VIF	1.535		1.430		1.535		1.612	
AR(1)	3.908***		6.748***		3.908***		4.102***	
AR(2)	0.224		0.240		0.224		0.235	
Sargan OIR test	0.321		0.346		0.321		0.337	
Hansen OIR test	0.277		0.265		0.211		0.291	
Observations	3,413		1,992		1,421		3,413	

*, **, *** represent statistical significance at 10%, 5%, and 1% levels.

Note. In Column 1, the regression results of ABN-NETHIRE are shown. In Columns 2 and 3, we re-estimate the baseline regression model using sub-samples of firms with labor overinvestment (positive abnormal net hiring—actual net hiring exceeds expected levels) in Column 2 or labor underinvestment (negative abnormal net hiring—actual net hiring below expected levels) in Column 3. In Column 4, we used the two-stage least squares approach to justify our main findings. ABN-NETHIRE = abnormal net hiring; AI = artificial intelligence; S/E = standard error; TMT = top management team; OIR = overidentifying restrictions; MTB = market to book; H1 = Hypothesis 1.

We included several control variables to account for factors that may affect AI adoption and labor investment efficiency. FIRM-SIZE, FIRM-AGE, MTB, DIVIDEND, and LABOR-INTENSITY are significantly associated with ABN-NETHIRE. Older businesses might have better labor management techniques due to their well-established procedures and acquired expertise, which improves labor investment efficiency. Businesses that have better market worth than their book value have better labor investment efficiency (Wu et al., 2022; Zhang et al., 2025). As investors believe these businesses have a better chance of growth, their workforce and other resources can be utilized more effectively. The beneficial effects of dividends highlight how shareholder expectations influence productivity (Le & Tran, 2022). Regular dividend payers may be more motivated to maximize labor investment to sustain profitability and meet investor demands (Le & Tran, 2022). Similarly, labor investment efficiency benefits from labor intensity, which measures the proportion of labor costs in a company's operational

activities. Firms with higher labor intensity are likely to be more concerned with optimizing the labor process, resulting in increased productivity.

In contrast, LEVERAGE, SD-SALES, SD-CASHFLOW, and SD-NETHIRE negatively affect labor investment efficiency. Leverage limits a firm's labor cost management flexibility, as debt repayments may take precedence over labor efficiency expenditures (Hossain et al., 2022). Sales and cash flow volatility with high standard deviations means that these affect labor planning and investments (Schopohl et al., 2024). Firms with high volatility may struggle to commit to long-term labor investment strategies, which causes inefficiency (Schopohl et al., 2024). SD-NETHIRE measures new hire variability, revealing that companies with unstable hiring processes may struggle to maximize labor investments. We find no evidence that tangible assets affect labor investment efficiency. Tangible assets are essential in operations, but they may not directly impact labor efficiency because they are more closely tied to capital-intensive activities than to labor optimization.

Last, TMT characteristics such as diversity and age help us understand organizational decision-making, performance, and strategy (Xu & Deshmukh, 2024). Likewise, racial diversity improves inclusive strategies and customer management, while different tenures bring new visions and knowledge (Semwal et al., 2024). As the members of TMT govern business rules, make important business decisions, and form the firm's vision, they affect how the firm adapts and seizes opportunities. Their backgrounds impact how a firm reacts to environmental stimuli in a dynamic competitive market (Xu & Deshmukh, 2024).

We included country, industry, year, and a Covid-19 dummy to control external stimuli that may impact the results and isolate the actual impact of TMT traits on ABN-NETHIRE. Country impacts account for economic, regulatory, and cultural differences between countries (Lin et al., 2021). Industry-specific variables, including market competition, technological advancements, and regulatory frameworks, influence performance. Year accounts for macroeconomic cycles, trends, and global events such as recessions and pandemics (Lin et al., 2021). To account for COVID-19's impact, this study set 1 for the years affected by the pandemic (2020–2021) and 0 otherwise.

Moderation Effect

We conducted further analysis to determine the effect of our moderation. We hypothesized that globalization moderates the relationship between AI adoption and labor investment efficiency. The direct impact of AI adoption on ABN-NETHIRE is statistically significant, indicating that AI adoption improves labor investment efficiency. Globalization measures, including KOF ($\beta = 0.048$, $p < 0.10$; see Table 4), DAI ($\beta = 0.037$, $p < 0.10$), GCI ($\beta = 0.057$, $p < 0.10$), and TFP ($\beta = 0.117$, $p < 0.05$), have positive and significant direct effects. Higher globalization is associated with improved labor investment efficiency because firms in these nations are more likely to embrace AI technology, which enhances productivity and workforce management (Abbas Khan et al., 2024; Elfaki & Ahmed, 2024).

Hypothesis 2 concerns the interaction between AI adoption and the globalization metrics. AI-ADOPTION * KOF ($\beta = 0.146$, $p < 0.01$), AI-ADOPTION * DAI ($\beta = 0.096$, $p < 0.05$), AI-ADOPTION * GCI ($\beta = 0.125$, $p < 0.01$), and AI-ADOPTION * TFP ($\beta = 0.168$, $p < 0.01$) interaction factors ($\beta = 0.146$, $p < 0.01$) exhibit significantly more significant coefficients than the independent variables' direct effects. This suggests that globalization strongly moderates AI usage and labor investment efficiency (Sheraz et al., 2021; Tingbani et al., 2025). AI adoption improves labor investment efficiency more in globalized countries (Sheraz et al., 2021). AI adoption benefits multinational environments more, where they may utilize stronger infrastructure. These companies are more technologically prepared due to the presence of global knowledge and practices (Kinkel et al., 2022). The higher coefficient for interaction terms suggests that the benefits of globalization in digital infrastructure, institutional support, or competitiveness strengthen AI usage's impact. High-KOF countries can use best global practices and networks to integrate AI in a more efficient manner and increase labor investment efficiency (Jain & Jindal, 2022).

Firms can leverage automation, data-driven decision-making, and other efficiency-boosting technologies to use AI in countries with strong digital infrastructure (Liu et al., 2023). Globalization influences the extent to which AI improves worker efficiency (Skare & Riberio Soriano, 2021). The *F*-statistics reveal notable differences in the direct and interaction effects of using AI. The significant differences in *F*-test coefficients indicate that globalization plays a moderating role in the relationship between using AI and labor investment efficiency, enhancing this effect. It also increases AI adoption and company performance. These results support that globalization, together with KOF, DAI, GCI, and TFP, improves AI usage. Highly globalized countries enhance technological including AI adoption, which further improves labor management and productivity. Globalization measures enhance technological infrastructure, access to global information, and competitive business settings significantly modify the relationship between AI adoption and labor efficiency (Huang & Liu, 2024; Skare & Riberio Soriano, 2021).

Table 4. Moderation Effect

Variables	Model 1		Model 2		Model 3		Model 4	
	ABN-NETHIRE		ABN-NETHIRE		ABN-NETHIRE		ABN-NETHIRE	
	Coefficient	S/E	Coefficient	S/E	Coefficient	S/E	Coefficient	S/E
AI-ADOPTION (H1)	0.032**	(0.015)	0.022**	(0.011)	0.019**	(0.009)	0.028**	(0.012)
One-year lagged								
(ABN-NETHIRE)	0.632***	(0.145)	0.617***	(0.140)	0.640***	(0.138)	0.632***	(0.098)
Moderator								
KOF	0.048*							
DAI			0.037*					
GCI					0.047*			
TFP							0.117**	
Interaction effect								
AI-ADOPTION * KOF	0.146***							
AI-ADOPTION * DAI			0.096**					
AI-ADOPTION * GCI					0.125***			
AI-ADOPTION * TFP							0.168***	
F-statistics	3.523***		4.999***		5.078***		6.889***	
Financial control								
FIRM-SIZE	0.082***	(0.017)	0.162***	(0.033)	0.149***	(0.029)	0.109***	(0.015)
FIRM-AGE	0.057**	(0.027)	0.076**	(0.036)	0.042**	(0.019)	0.116**	(0.037)
LEVERAGE	-0.104**	-(0.048)	-0.124**	(0.056)	-0.084**	(0.036)	-0.144**	(0.045)
MTB	0.063*	(0.037)	0.093*	(0.054)	0.097*	(0.053)	0.091*	(0.036)
SD-SALES	-0.113**	(0.050)	-0.109**	(0.048)	-0.113**	(0.047)	-0.114**	(0.034)
SD-CASH FLOW	-0.128***	(0.024)	-0.124***	(0.023)	-0.128***	(0.022)	-0.129***	(0.016)
SD-NETHIRE	-0.022**	(0.010)	-0.021**	(0.010)	-0.022**	(0.010)	-0.020**	(0.006)
TANGIBLES	0.002	(0.002)	0.002	(0.002)	0.002	(0.002)	0.002	(0.002)
DIVIDEND	0.074**	(0.314)	0.072**	(0.302)	0.074**	(0.295)	0.075**	(0.215)
LABOR-INTENSITY	0.067*	(0.036)	0.065*	(0.035)	0.067*	(0.034)	0.068**	(0.025)
TMT control								

continued on following page

Table 4. Continued

Variables	Model 1		Model 2		Model 3		Model 4	
	ABN-NETHIRE		ABN-NETHIRE		ABN-NETHIRE		ABN-NETHIRE	
	Coefficient	S/E	Coefficient	S/E	Coefficient	S/E	Coefficient	S/E
TMT age	0.001	(0.001)	0.001	(0.001)	0.001	(0.001)	0.001	(0.001)
TMT diversity	0.033*	(0.018)	0.032*	(0.017)	0.048**	(0.017)	0.033*	(0.012)
TMT race	0.082**	(0.035)	0.079**	(0.034)	0.082**	(0.033)	0.083**	(0.024)
TMT tenure	0.093**	(0.045)	0.090**	(0.043)	0.093**	(0.042)	0.094**	(0.031)
Fixed control								
Country fixed effect	Yes		Yes		Yes		Yes	
Year fixed effect	Yes		Yes		Yes		Yes	
Industry effect	Yes		Yes		Yes		Yes	
Covid-19 effect	Yes		Yes		Yes		Yes	
Weighted statistics								
$R-squared$	0.481		0.513		0.481		0.505	
*Adjusted R*squared	0.465		0.497		0.465		0.488	
Regressions' S/E	0.222		0.237		0.222		0.233	
F-statistics	139.496		149.115		139.496		146.423	
Probability (F-statistics)	0.000		0.000		0.000		0.000	
Mean VIF	1.402		1.306		1.402		1.473	
AR(1)	3.570***		6.166***		3.570***		3.748***	
AR(2)	0.205		0.220		0.205		0.215	
Sargan OIR test	0.293		0.316		0.293		0.308	
Hansen OIR test	0.253		0.242		0.253		0.265	
Observations	3,413		3,413		3,413		3,413	

*, **, *** represent statistical significance at 10%, 5%, and 1% levels.
Note. In this table, we report the results of the moderation effects. We used KOF, DAI, GCI, and

TFP measures of globalization. These measures have a high level of correlation (see Table 6 in the Appendix). We used separate regressions for each of the globalization constructs. The control factors are the same as those used in our primary regression. The results of the control factors are not explained to avoid repetitiveness. ABN-NETHIRE = abnormal net hiring; AI = artificial intelligence; S/E = standard error; TMT = top management team; OIR = overidentifying restrictions; VIF = variance inflation factor; KOF = Globalization Index; DAI = Digital Adoption Index; GCI = Global Competitiveness Index; TFP = total factor productivity; MTB = market to book; H1 = Hypothesis 1; SD = standard deviation; SD-NETHIRE = standard deviation of employee volatility.

Additional Test

Alternative Measures of Labor Investment Efficiency

This study explores different indicators to investigate labor investment sensitivity efficiency estimates using diverse proxies. It applies the findings to different labor efficiency concepts. It begins with Cao and Rees (2020), utilizing the industry median net hiring as the optimal labor investment level for a sector. This method allows us to compare each firm's labor investment efficiency to industry standards, identifying differences that may suggest labor

misallocation. Second, using the idea from Biddle et al. (2009)'s firm-specific model of investment as a function of sales growth, this paper considers labor investment and variances by concentrating on model residuals. Firm labor allocation inefficiencies are reflected in these residuals, indicating a misalignment between labor investment and economic fundamentals. We also incorporate capital investment, research and development expenses, acquisition costs, and the logarithm of gross domestic product per capita into Pinnuck and Lillis's (2007) model. This enlarged model incorporates macroeconomic elements that may influence firms' labor decisions in ways that previous models cannot, offering a more comprehensive view of how nonlabor investments impact labor efficiency (Khedmati et al., 2020). As seen in Table 5, our thorough checks show that our major conclusions are consistent with any labor investment efficiency proxy. This strengthens our findings and analysis, making our insights on labor investment efficiency broad and credible.

Table 5. Alternative Proxies of Abnormal Net Hiring (ABN-NETHIRE)

	Cella (2020)	**Biddle et al. (2009)**	**Pinnuck and Lillis (2007)**
			Augmented
AI-ADOPTION	0.040**	0.033**	0.029**
One-year lagged			
(ABN-NETHIRE)	0.616***	0.627***	0.588***
Control factors	Yes	Yes	Yes
Fixed control	Yes	Yes	Yes
Country fixed effect	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes
Industry effect	Yes	Yes	Yes
Weighted statistics			
$R-squared$	0.465	0.439	0.496
Adjusted R squared	0.450	0.424	0.481
Regressions' S/E	0.215	0.203	0.229
F-statistics	134.883	127.253	144.184
Probability (*F*-statistics)	0.000	0.000	0.000
Mean VIF	1.356	1.279	1.263
AR(1)	4.394***	6.111***	3.060***
AR(2)	0.198	0.187	0.213
Sargan OIR test	0.283	0.267	0.306
Hansen OIR test	0.245	0.231	0.234

, **, *** denote 5%, and 1% statistical significance.

Note. Regressing alternative abnormal net hiring measures on employee treatment yields this table. Cella (2020)'s alternative anomalous net hiring regression results are in Column 1. We estimate a firm's ideal net hiring level using its industry median. Column 2 shows results from Biddle et al. (2009) on alternative abnormal net hiring regression. We estimate a firm's labor investment model as a function of sales growth and utilize residual absolute value as a surrogate for labor investment variations. The alternative anomalous net hiring regression using the augmented Pinnuck and Lillis (2007) model is shown in Column 3. We include capital expenditure, research and development spending, acquisition expenses, and the gross domestic product per capita logarithm to reestimate the model. Parentheses indicate robust standard errors clustered at the firm level for *p*-values. ABN-NETHIRE = abnormal net hiring; AI = artificial intelligence; S/E = standard error; OIR = overidentifying restrictions; VIF = variance inflation factor.

DISCUSSION

This research has practical implications for enterprises, governments, and economic planners regarding digitization. It indicates that AI enhances labor investment efficiency, especially in key GCC industries such as finance, energy, and logistics. AI helps firms optimize labor management, which is particularly important in cities with large migrant populations. Adopting AI early allows firms to compete internationally by raising productivity, cutting expenses, and expanding strategy. Governments should use AI for workforce development and digital transformation.

Governments worldwide should provide relevant education, training, technology, and regulatory frameworks according to the city's digital maturity and economic development direction. They should encourage research and provide related grants to develop hardware and software that can meet the needs of both industry and society. For example, specialized robots may be beneficial to the manufacturing industry in the production of automobiles and the laying of bricks on construction sites. A lack of digital infrastructure or skills may prevent less globalized enterprises from utilizing AI. Places such as the GCC and others with a high proportion of migrant laborers require specific AI policies to meet the industries' needs. To compete in a globalized environment, policymakers and businesses may also use AI for staff recruitment and management.

CONCLUSION

This research examines the impact of globalization on the effects of AI applications, which in turn influence firms' labor investment efficiency. It confirms that AI enhances labor investment efficiency through automation, improves decision-making with data, and optimizes staff allocation across different departments. Firms use AI to enhance human resource management, productivity, and performance. Globalization raises competition, enhances technology, and promotes market integration worldwide, thereby magnifying AI's impact on labor investment efficiency. This research contributes to academic scholarship on globalization as a moderating factor in AI's impact on firms' workforce planning and management. It supports dynamic capabilities theory, which posits that AI enables businesses to adapt to market changes.

This study lays a solid ground and rationale for firms to invest in and adopt AI. As different industries have varying levels of work complexity, some may be more repetitious, while others require more innovative work. Future studies may compare different industries, such as the creative industries and the construction sector, that adopt AI and examine its impact on labor investment efficiency. Second, future studies may consider government AI strategic policies, regulations, privacy laws, education systems, and the implications of digital infrastructure on the influence of AI on labor productivity. Finally, a cross-country and cultural comparative analysis may help us understand how culture influences labor investment efficiency. All these will help enhance generalizability and refine the existing theories.

CONFLICTS OF INTEREST

We wish to confirm that there are no known conflicts of interest associated with this publication.

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APPENDIX

Table 5. Description of Variables

Variable	Description
NETHIRE	Current year employees minus last year's employees, scaled by last year's employees
FIRM-SIZE	Natural log of firm *i*'s total assets in year *t*
FIRM-AGE	The difference between the year of observation and the year of incorporation of firm *i* in year *t*
LEVERAGE	Debt-to-equity ratio of firm *i* in year *t*
MTB	Firm *i* market to book value in year *t*
SD-SALES	Standard deviation of sales of firm *i* in year *t*
SD-CASH FLOW	Standard deviation of cash flows of firm *i* in year *t*
SD-NETHIRE	Standard deviation of NETHIRE of firm *i* in year *t*
TANGIBLES	Firm *i*'s tangible assets scaled by total assets
DIVIDEND	Dummy variable equal to 1 if firm *i* pays a dividend in year *t*, and 0 otherwise
LABOR-INTENSITY	The number of employees divided by total assets at the end of year *t* for firm *i*
TMT age	Average of total members' age in the TMT of firm *i* in year *t*
TMT diversity	The ratio of female members in the TMT of firm *i* in year *t*
TMT race	Percentage of foreign directors in the TMT
TMT tenure	Average of total TMT members' tenure of firm *i* in year *t*

Note. TMT = top management team.

Table 6. Globalization Index Correlation Matrix

Variables	KOF	DAI	GCI	TFP
KOF	1.000			
DAI	0.836***	1.000		
GCI	0.895***	0.780***	1.000	
TFP	0.794***	0.816***	0.765***	1.000

*** denotes significance at a 1% level.

Note. The results of the Pearson correlation are presented here. The variables are Globalization Index (KOF), Digital Adoption Index (DAI), Global Competitiveness Index (GCI), and total factor productivity (TFP).